

CENTENNIAL, CO, USA

WMO: 724666

Lat: 39.570N

Lon: 104.849W

Elev: 5883

StdP: 11.83

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93067

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.9	7.1	-7.8	4.5	15.3	-3.5	5.7	23.1	27.2	41.8	23.9	39.3	5.2	40	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	92.2	59.3	90.0	59.0	87.5	58.7	64.7	80.0	63.1	78.3	61.8	77.5	9.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.9	99.6	68.1	58.6	91.8	67.7	56.7	85.6	67.2	33.0	79.4	31.6	77.3	30.5	77.1	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.5	20.5	18.1	-7.0	97.3	4.3	2.0	-10.1	98.8	-12.6	99.9	-15.0	101.0	-18.2	102.5		
(5)			-7.6	66.8	4.3	2.2	-10.7	68.4	-13.2	69.7	-15.6	70.9	-18.7	72.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	32.2	33.2	41.0	46.7	55.9	67.1	73.4	71.1	63.4	50.4	40.3	31.8
	DBStd	17.32	10.55	11.11	10.35	9.48	8.66	6.95	4.86	4.80	8.12	9.69	10.59	10.98
	HDD50	2552	554	473	308	168	44	1	0	0	11	112	314	567
	HDD65	5915	1017	889	745	549	297	58	5	10	120	455	740	1030
	CDD50	2788	2	4	28	70	228	513	724	656	412	125	24	2
	CDD65	674	0	0	0	1	16	119	264	201	70	3	0	0
	CDH74	8059	0	0	4	25	323	1647	2901	2069	974	115	1	0
	CDH80	3173	0	0	0	1	69	659	1322	810	302	10	0	0
	Wind	WSAvg	8.3	7.9	8.5	9.0	9.7	8.7	8.2	7.9	7.7	7.8	7.7	7.7
	Precipitation	PrecAvg	16.9	0.6	0.7	1.3	2.2	2.3	1.9	2.2	2.1	1.4	1.1	0.8
(15)		PrecMax	24.1	1.3	2.2	4.3	5.5	5.9	4.4	4.4	4.5	6.9	2.8	2.8
(16)		PrecMin	9.8	0.1	0.0	0.0	0.1	0.6	0.3	0.3	0.0	0.0	0.2	0.0
(17)		PrecStd	3.4	0.3	0.5	1.0	1.2	1.3	1.1	1.2	1.3	0.7	0.6	0.6
(18)														
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.0	66.6	73.4	77.1	86.3	94.1	96.7	93.3	90.9	81.4	72.5	63.8
		MCWB	42.3	43.5	47.9	49.3	55.2	58.4	60.6	59.2	58.2	52.6	46.3	42.4
	2%	DB	58.0	61.1	68.6	73.1	81.4	90.4	93.0	90.4	86.4	77.4	67.2	58.0
		MCWB	39.5	41.1	45.0	48.1	54.3	57.8	59.8	59.0	56.3	50.5	44.5	39.8
	5%	DB	53.4	56.1	64.1	69.9	77.5	87.6	90.6	87.9	83.6	73.1	63.2	53.6
		MCWB	37.7	38.8	43.4	46.8	52.9	57.3	59.7	59.0	55.4	49.3	43.0	37.8
	10%	DB	48.4	51.3	59.8	65.2	73.2	83.8	88.0	84.8	80.5	68.4	58.0	48.5
		MCWB	35.5	36.7	41.4	45.3	52.1	56.9	59.6	59.1	55.0	47.9	40.8	35.5
	0.4%	WB	43.8	44.6	50.8	52.3	59.8	64.9	67.2	66.9	63.1	56.3	47.5	43.3
		MCDB	60.6	64.0	67.1	69.9	78.4	82.4	83.5	81.6	79.4	72.3	67.9	62.0
(27)	2%	WB	40.4	41.7	46.3	50.0	57.1	62.2	65.0	64.5	60.2	53.1	45.7	40.4
(28)		MCDB	56.2	59.5	65.7	67.7	73.6	79.0	80.5	79.0	76.7	70.5	66.1	56.8
(29)	5%	WB	38.0	39.3	44.0	48.1	55.2	60.3	63.4	62.9	58.4	50.8	43.5	38.1
(30)		MCDB	52.0	54.8	62.4	65.9	71.5	77.6	78.4	77.6	75.0	68.6	61.8	52.4
(31)	10%	WB	35.8	37.0	41.8	46.2	53.3	59.0	62.1	61.5	56.9	48.9	41.3	35.7
(32)		MCDB	48.1	50.4	58.1	63.4	69.3	76.7	77.8	76.5	73.6	65.7	56.9	48.0
(33)	Mean Daily Temperature Range	MDBR	23.5	23.6	25.0	24.1	24.4	27.0	26.4	25.5	26.4	25.6	24.7	23.3
(34)		5% DB	28.8	30.2	31.0	31.1	30.5	32.0	30.6	29.4	30.9	32.4	31.3	29.2
(35)		MCWBR	15.5	15.7	15.1	13.8	11.8	10.8	9.3	8.9	10.3	13.5	15.3	15.9
(36)		5% WB	27.6	28.8	29.8	28.4	27.2	27.4	25.3	25.0	26.7	28.2	30.0	28.1
(37)		MCWBR	15.4	15.6	15.2	13.6	12.1	11.2	9.5	9.4	10.0	12.6	15.2	15.8
(38)	Clear-Sky Solar Irradiance	taub	0.236	0.242	0.263	0.291	0.310	0.319	0.347	0.337	0.301	0.267	0.247	0.238
(39)		taud	2.632	2.626	2.572	2.507	2.515	2.503	2.466	2.480	2.560	2.643	2.622	2.624
(40)		Ebn at Noon	308	319	319	313	307	303	293	294	301	305	300	299
(41)		Edh at Noon	21	24	29	33	33	34	35	33	29	23	21	19
(42)	All-Sky Solar Radiation	RadAvg	829	1144	1526	1808	1997	2265	2144	1926	1672	1240	892	726
(43)		RadStd	35	59	116	124	155	105	96	66	87	75	49	39

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			+0.56	N/A	N/A	N/A	N/A	N/A	N/A	+126	+61
(47) Regional (28 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	N/A

Nomenclature: See separate page